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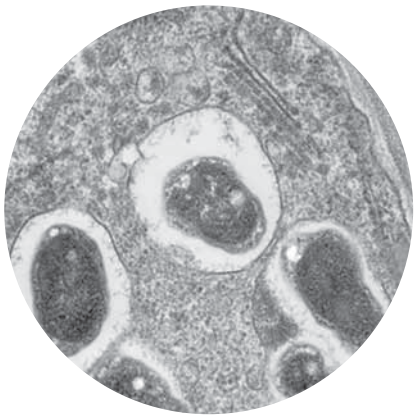


**Figure 12.6:** Rhizobium (Bacteroid) in root nodule

4. A membrane bound bacterium is formed inside the nodule and is called **bacteroid**.
5. Cytokinin from bacteria and auxin from host plant promotes cell division and leads to nodule formation

**Activity**

- Collect roots of legumes with root nodules.
- Take cross section of the root nodule.
- Observe under microscope. Discuss your observations with your teacher.

### Non-Legume

*Alnus and Casuarina* contain the bacterium *Frankia*. *Psychotria* contains the bacterium *Klebsiella*.

**ii. Nitrogen fixation without nodulation**

**The following plants and prokaryotes are involved in nitrogen fixation.**

Lichens - *Anabaena* and *Nostoc*

*Anthoceros* - *Nostoc*

*Azolla* - *Anabaena azollae*

*Cycas* - *Anabaena* and *Nostoc*

### b. Non-symbiotic Nitrogen fixation

Free living bacteria and fungi also fix atmospheric nitrogen.

| | |
|---|---|
| Aerobic | *Azotobacter, Beijerneckia and Derxia* |
| Anaerobic | *Clostridium* |
| Photosynthetic | *Chlorobium and Rhodospirillum* |
| Chemosynthetic | *Disulfovibrio* |
| Free living fungi | Yeast and *Pullularia* |
| Cyanobacteria | *Nostoc, Anabaena and Oscillatoria.* |

## 12.8 Nitrogen cycle and nitrogen metabolism

### 12.8.1 Nitrogen cycle

CNFN3G

This cycle consists of following stages:

#### 1. Fixation of atmospheric nitrogen

Di-nitrogen molecule from the atmosphere progressively gets reduced by addition of a pair of hydrogen atoms. Triple bond between two nitrogen atoms (N≡N) are cleaved to produce ammonia (Figure 12.7).

nitrogen fixation process requires Nitrogenase enzyme complex, Minerals (Mo, Fe and S), anaerobic condition, ATP, electron and glucose 6 phosphate as $H^+$ donor. Nitrogenase enzyme is active only in anaerobic condition. To create this anaerobic condition a pigment known as **leghaemoglobin** is synthesized in the nodules which acts as oxygen scavenger and removes the oxygen. Nitrogen fixing bacteria in root nodules appears pinkish due to the presence of this leghaemoglobin pigment.